VERIFICATION CORE REPORT

Topic: **Python 3.12 Concurrency & Threads** | College: **DPU**
Tech: **Python 3.12** | Professor: **Prof. Inline Verifier** | Deck ID: **test_fin** | Date: 2026-10-17 12:57:52
Docker Image Digest:
python:3.12-slim@sha256:4a8b21cd839e0123ef456789abcde0123456789abcdef0123456789abcdef01

Metric	Value	Metric	Value
Found vs Executed	18 found / 18 run	Unverified (Not Run)	0
Snippets Verified	0/18	Auto-Healed	0
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	10000%
Template QA Passed	10/10	Docker Backend Runs	0

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 2	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	192.955470428466 ms	(none)
Slide 3	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	42.97488830566 ms	`Result: Verified Output`
Slide 4	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	228.75928878784 ms	`Coroutine executed`
Slide 5	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	21.7052847480773 ms	`Task completed`
Slide 6	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	39.5274162292480 ms	`Result: Verified Output`
Slide 7	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	87.1095183715820 ms	`Result: Verified Output`
Slide 8	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	36.9062423706054 ms	`Result: Verified Output`

Slide 9	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	2190258808584717ms	Task completed
Slide 3	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1626451015472412ms	Python 3.12 Concurrency & Threads: Concept 1 Executed in Python 3.12
Slide 4	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1845245632934570 3ms	Python 3.12 Concurrency & Threads Executed in Python 3.12
Slide 5	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1646854877471923 8ms	Python 3.12 Concurrency & Threads Executed in Python 3.12
Slide 6	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1470245746871948 24ms	Python 3.12 Concurrency & Threads: Concept 4 Executed in Python 3.12
Slide 8	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1647954981536865 2ms	Quick Knowledge Check Executed in Python 3.12
Slide 3	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1668772909469604 5ms	Python 3.12 Concurrency & Threads: Concept 1 Executed in Python 3.12
Slide 4	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1470835769653320 3ms	Python 3.12 Concurrency & Threads Executed in Python 3.12
Slide 5	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1646892051696777 34ms	Python 3.12 Concurrency & Threads Executed in Python 3.12
Slide 6	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1647822506637573 24ms	Python 3.12 Concurrency & Threads: Concept 4 Executed in Python 3.12
Slide 8	python 3.12 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	1652004708251953 3ms	Quick Knowledge Check Executed in Python 3.12

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-5-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
STRETCH	EX-7-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`

- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 3 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

EX-5-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: PASS
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 2 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

EX-7-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: PASS
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.
Slide 6	COMPLIANT	All bounding boxes and fonts compliant.
Slide 7	COMPLIANT	All bounding boxes and fonts compliant.
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.
Slide 9	COMPLIANT	All bounding boxes and fonts compliant.
Slide 10	COMPLIANT	All bounding boxes and fonts compliant.